# Toward the Dark Factory

## Agentic Engineering, Harness Design, and Securing the Pipeline

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# Dark Factory

**Dark Factory /** 無人工場 (mujin kōjō)
Lights-off, fully automated, literally "no humans factory"

- Emerged in **1980s** as a future vision for manufacturing (FANUC 1981, GE 1982)
- FANUC became the canonical lights-out factory in 2001

Now in the **2020s** this percolating for software.

StrongDM's Dark Factory thesis:

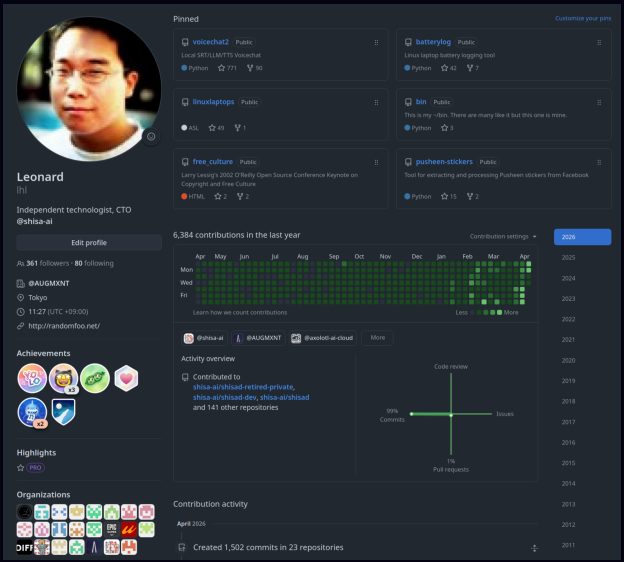
> Code must not be written by humans.
> Code must not be reviewed by humans.

At OpenAI, Ryan Lopopolo's team is building Codex and talking about **harness engineering**:

- **5 months, 1M+ lines of code, zero human-written code, zero pre-merge human review. ~$2-3K/day in tokens**

自働化 (jidoka) - autonomation
automation with a human touch (Toyota)

# Ex-Coder


Pinned
Customize your pins
voicechat2 Public
Local SRTL LLM/TTS Voicechat
Python 771 90
batterylog Public
Linux laptop battery logging tool
Python 42 7
linuxlaptops Public
ASL 49 1
bin Public
This is my ~/bin. There are many like it but this one is mine.
Python 3
free_culture Public
Larry Lessig's 2002 O'Reilly Open Source Conference Keynote on Copyright and Free Culture
HTML 2 2
pusheen-stickers Public
Tool for extracting and processing Pusheen stickers from Facebook
Python 15 2
Leonard
lhl
Independent technologist, CTO
@shisa-ai
Edit profile
361 followers · 80 following
@AUGMXNT
Tokyo
11:27 (UTC +09:00)
http://randomfoo.net/
Achievements
x3
x2
Highlights
PRO
Organizations
6,384 contributions in the last year
Contribution settings
2026
2025
2024
2023
2022
2021
2020
2019
2018
2017
2016
2015
2014
2013
2012
2011
Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr
Mon
Wed
Fri
Learn how we count contributions
Less More
@shisa-ai @AUGMXNT @axolotl-ai-cloud More
Activity overview
Contributed to
shisa-ai/shisad-retired-private,
shisa-ai/shisad-dev, shisa-ai/shisad
and 141 other repositories
Code review
99%
Commits
Issues
1%
Pull requests
Contribution activity
April 2026
Created 1,502 commits in 23 repositories

# Token Count

**1 month: March 11 - April 11:**

| Provider | Input | Output | Reasoning | Cache Read | Total | Cost |
|---|---|---|---|---|---|---|
| Codex | 434M | 41M | 23M | 7.92B | 8.39B | $3,423 |
| Claude | 23M | 12M | 420M | 3.54B | 3.99B | $4,202 |
| **Total** | **457M** | **53M** | **443M** | **11.46B** | **12.38B** | **$7,625** |

**92.5% cache read** — 94.3% Codex, 88.6% Claude

```
>_ OpenAI Codex (v0.118.0)

Visit https://chatgpt.com/codex/settings/usage for up-to-date
information on rate limits and credits

 Model:                       gpt-5.4 (reasoning xhigh, summaries auto)
 Directory:                   ~/sansho
 Permissions:                 Custom (workspace-write, on-request)
 Agents.md:                   AGENTS.md
 Account:                     lni@randomfoo.net (Pro)
 Collaboration mode:          Default
 Session:                     019d8eda-ad46-7df3-87f8-c78ce086f838

 5h limit:                    [                    ] 100% left (resets 15:56)
 Weekly limit:                [                    ] 0% left (resets 01:54 on 17 Apr)
 GPT-5.3-Codex-Spark limit:
 5h limit:                    [                    ] 100% left (resets 15:56)
 Weekly limit:                [                    ] 97% left (resets 00:05 on 19 Apr)
```

```
Overview   Models

    Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr
Mon
Wed
Fri

    Less  More

All time · Last 7 days · Last 30 days

Favorite model: Opus 4.6          Total tokens: 13.4m

Sessions: 135                     Longest session: 6d 9h 9m
Active days: 25/30                Longest streak: 11 days
Most active day: Mar 14           Current streak: 0 days

You've used ~18x more tokens than War and Peace

 ↓ stats · r to cycle dates · ctrl+s to copy
```

# Aside: Unstupify Claude

```
// ~/.claude/settings.json
{
  "cleanupPeriodDays": 99999,
  "env": {
    "CLAUDE_CODE_EFFORT_LEVEL": "max",
    "CLAUDE_CODE_DISABLE_ADAPTIVE_THINKING": "1",
    "MAX_THINKING_TOKENS": "128000",
    "CLAUDE_CODE_SUBAGENT_MODEL": "sonnet"
  },
  "attribution": {
    "commit": "",
    "pr": "" },
  "permissions": {
    "defaultMode": "bypassPermissions"
  },
  "skipDangerousModePermissionPrompt": true
}
```

# AgentsView

https://github.com/wesm/agentsview (https://github.com/wesm/agentsview)

# The Migration Path

```
Manual → Chat → Autocomplete → Agent-driven → Dark factory
                  (Copilot)          ↑
                                    HERE
```

- **Agent-driven** = agents write the code, you govern the process
- **Dark factory** = agents autonomously plan, implement, test, review, ship

# My Current Setup

- byobu (tmux) + aggressive pane labeling — one session per project
- Claude Code, Codex, Amp, OpenCode, pi — use the best for the job
- Remote via Tailscale

See Prashant's **Agentic Coding Weekly #14** (https://www.agenticcodingweekly.com/p/acw-14-claude-sonnet-4-6-gemini-3-1-pro-and-spec-driven-development) for a February writeup (8B cached tokens/month)

## ▒▒▒ Model Roles (April 2026)

| Role | Model | Why |
|---|---|---|
| **Coder** | GPT-5.4 xhigh | Best general coder for my workload |
| **Planner/Writer** | Opus 4.6 max | Best for interactive planning, writing |
| **Reviewer** | Multi-model (4 lanes) | R1: Opus, R2: 5.2, R3: 5.3-Codex, R4: 5.4 |

# Pareto Efficiency, Not FOMO

> "There's a lot of churn and FOMO in the space, but things get 'baked-in' very quickly. My goal is Pareto efficiency: adopt what gives the best gains for my actual work."

**What actually matters (the Pareto 20%):**

1. A **spec-driven dev loop** (any form — AGENTS.md, GSD phases, doesn't matter)
2. **Test-first development**
3. **Independent multi-model review**
4. Your harness evolving with your project, not with the plugin ecosystem

# Currently Working On

| Project | What | Speed/LOC |
|---|---|---|
| **shisad** | Security-first AI agent daemon | 320K LOC, v0.1→v0.6.4 |
| **textguard** | Prompt injection detection | **6.5 hours**, 4.4K LOC |
| **outline-edit** | Outline KB CLI + local cache | **3 hours**, 2.2K LOC |
| **chotto** | Translation fine-tuning | autoresearch |
| **shisa-asr** | Audio fine-tuning | autoresearch, 31-iter sweeps |
| **sansho** | Model pretraining | autoresearch, hybrid looping arch |
| **vLLM optims** | Attention kernels | autoresearch, +240% throughput |
| **kvcache-quant** | kvcache research | autoresearch, >25X compression |
| **specdec** | diffusion-based | autoresearch, >2X throughput |

# shisad

https://github.com/shisa-ai/shisad

shisad public tree:

| Language | Files | Lines | Code | Complexity |
|---|---|---|---|---|
| Python | 398 | 138,791 | 121,980 | 10,967 |
| Markdown | 29 | 9,500 | 7,448 | 0 |
| JSON | 14 | 343 | 343 | 0 |
| Shell | 6 | 1,018 | 805 | 174 |
| YAML | 3 | 519 | 428 | 0 |
| TOML | 1 | 103 | 92 | 0 |
| **Total** | **453** | **150,476** | **131,266** | **11,141** |

Estimated Cost to Develop (organic) **$4,525,381**
Estimated Schedule Effort (organic) **24.41 months**
Estimated People Required (organic) **16.47**

# shisad: Automation

| Metric | v0.6.2 (human) | v0.6.3 (auto v1) | v0.6.4 (auto v2) |
|---|---|---|---|
| Wall-clock | **105.67h** | **38.85h** | **8.75h** |
| Agent session time | 27.8h | 70.9h | 21.2h |
| Wall/session ratio | 3.80 | 0.55 | **0.41** |
| Concurrent agents | 0.26 | 1.83 | **2.42** |
| Commits | 38 | 64 | 22 |
| Insertions | +16,009 | +7,414 | +1,378 |
| Deletions | -1,005 | -408 | -1,264 |
| Net lines | +15,004 | +7,006 | +114 |
| Review green rate | — | 17.6% | **42.2%** |
| ReleaseClose runs | — | 12 | **1** |

- **9.2x automation ratio:** wall/session went from 3.80 → 0.41 (human bottleneck → agent saturation)
- **9.3x more concurrent agents:** 0.26 → 2.42
- **First-pass green: 1 ReleaseClose run** (was 12 in v0.6.3)
- v0.6.4 **shrank** `src/` while growing `tests/` — faster AND cleaner

**DEMO**

# $implement

```
$implement v0.6.4
      │
  ├── CODER (GPT-5.4, autonomous)
  │   implements milestones from punchlist
  │
  ├── TARGETED VALIDATION (proportional tests)
  │
  ├── REVIEW FANOUT ($shisad-review)
  │   4 parallel reviewers:
  │   R1: Claude Opus 4.6   │  R2: GPT-5.2
  │   R3: GPT-5.3-codex     │  R4: GPT-5.4
  │
  ├── COLLATE-FIX-RESUBMIT
  │   batch findings, fix in one pass,
  │   re-review only non-green lanes
  │
  └── RELEASE CLOSE
      preflight → full 4-lane green gate → publish
```

# Reviewer Performance

**v0.6.3 → v0.6.4 comparison** (68+45 = 113 review runs, 249+113 = 362 lanes)

| Reviewer | Model | v0.6.3 | v0.6.4 | v0.6.3 min/i | v0.6.4 min/i |
|---|---|---|---|---|---|
| R1 | Opus 4.6 | 17.7% | **0.0%** | 39.4 | — |
| R2 | GPT-5.2 | 16.7% | 17.4% | 73.7 | 52.5 |
| R3 | GPT-5.3-codex | 18.0% | **32.1%** | 28.5 | **15.2** |
| **R4** | **GPT-5.4** | **59.1%** | **55.8%** | **9.1** | **10.0** |

**Combined across both releases (362 lanes):**

| Reviewer | Lanes | Issues | Issue Rate | Min/Issue | % of All Issues |
|---|---|---|---|---|---|
| R1 | 81 | 11 | 13.6% | 49.0 | 10.2% |
| R2 | 83 | 14 | 16.9% | 67.7 | 13.0% |
| R3 | 89 | 20 | 22.5% | 22.5 | 18.5% |
| **R4** | **109** | **63** | **57.8%** | **9.5** | **58.3%** |

Each reviewer catches **different classes of issues**

# Mining Your Own Process

When your dev process generates structured logs, analyze them.

**Tracked per release:**

- Wall-clock efficiency (commits/hour, insertions/hour)
- Session-time efficiency (active minutes, concurrent agents)
- Token efficiency (tokens per commit, tokens per net line)
- Review effectiveness (green rate, finding rates per reviewer)

**Others are doing this too:**

AMD's AI Group director data-mined **17,871 thinking blocks** and **234,760 tool calls** across 6,852 Claude Code sessions to demonstrate quality regression correlated with thinking-depth reduction. (anthropics/claude-code#42796)

# Autoloop Pattern

Karpathy-style autoresearch: agents run **autonomous experiments** overnight.

```
Define metric → Modify → Verify → Improved? Keep : Revert → Log →
Repeat
```

## ▒▒▒ Some of this week's Autoloop Results

- **PoPE**: 17 experiments, block_dsa_128 retained at 3.5X
- **Audio fine-tuning**: 31-iteration sweep, CWD broke 16-iteration SPREDS plateau
- **Tokenizer TX1**: Solved OOM (124 GiB → 81 GiB), built 128K vocab multi-lingual tokenizer (SuperBPE)
- **YOCO-U training**: 2.3x faster than dense baseline, 1.81x DDP scaling

# Harness Engineering

Ryan Lopopolo at OpenAI coined the term:

> "To let the model cook, you have to step back — take a systems thinking mindset, constantly asking where the agent makes mistakes, where you're spending time, and how to stop spending that time."

His key principles:

- **Decomposition:** Break down what the model can't do, build smaller blocks, reassemble
- **Encoding engineering taste:** Docs, tests, and review agents inject "what good looks like"
- **Signal harvesting:** Every failed build means the agent was missing context. Feed it back.
- **Build-time ratchets:** <1 minute for the full build — forces discipline

Invest in the harness, not the code.

# Spec-Driven Loop

```
RESEARCH/ANALYSIS
        │
     PLAN
        │
  IMPLEMENTATION (punchlist + worklog)  ◄── core control point
        │
   ┌── TEST (write tests first)
   ├── CODE (first pass, full punchlist)
   ├── REVIEW (independent, multi-model) ──┐
   ├── REMEDIATION ◄──────────────────────┘
   │   (repeat until green)
   └── ANALYSIS (post-milestone drift check)
        │
     STATUS (post-sprint review)
```

# Harness Spectrum

| Level | Example | What It Governs |
|---|---|---|
| **Lightweight** | AGENTS.md (~250 lines) | Dev loop, commit standards |
| **Process** | Superpowers plugin | TDD gates, subagent orchestration |
| **Governance** | shisad-dev $implement | Claim integrity, review traces |
| **Orchestration** | OpenAI Symphony | Multi-Agent automated dispatch, retry |
| **Autonomous** | StrongDM Attractor | LLM-as-judge, no human review |

For dark factory you need the **full stack** — process mechanics, governance, orchestration, and independent validation.

Right now, you will probably end up with Gastown, however.

# Supply Chain-pocalyps (March–April 2026)

**Trivy** (Mar 19–22) — most used OSS vuln scanner in CI/CD

- Force-pushed 75/76 tags to malicious commits
- Harvested SSH keys, cloud creds, K8s tokens from runner memory

**LiteLLM** (Mar 24) — PyPI creds stolen from CI that ran Trivy

- `.pth` file executes on every Python process startup
- 95M monthly downloads. Live for 3-5 hours.

**axios** (Mar 31) — lead maintainer's npm account compromised

- Published via stolen long-lived token, bypassing OIDC trusted publishing
- 100M weekly downloads. Live for ~3 hours.

# The Cascading Trust Problem

**One breach cascaded into dozens.**

```
Trivy compromised
    │
    ├──→ stole LiteLLM's PyPI creds ──→ LiteLLM compromised
    │
    └──→ CanisterWorm ──→ 47+ npm packages via stolen tokens
```

This is only going to get worse!

# 7-Day Rule

All three attacks had windows of hours. A 7-day policy trivially avoids them.

```
# npm (~/.npmrc)
min-release-age=7
ignore-scripts=true

# pnpm (pnpm-workspace.yaml)
minimumReleaseAge: 10080

# Python (uv)
UV_EXCLUDE_NEWER=$(date -d '7 days ago' -u +%Y-%m-%dT00:00:00Z)

# Bun (bunfig.toml)
[install]
minimumReleaseAge = 604800

# GitHub Actions: pin to full commit SHAs, not tags
- uses: actions/checkout@11bd71901bbe5b1630ceea73d27597364c9af683
```

**Supply Chain Security for Developers** (https://gist.github.com/lhl/f171eaea45df31a0b9287d7bf380657a)

# shisad: Secure Supply Chain & Publish Pipeline

If you publish or maintain open source software:

- **OIDC trusted publishing**
  - no long-lived PyPI/npm tokens
- **SBOM generation**
  - full dependency inventory on every release
- **pip-audit**
  - automated vulnerability scanning of all dependencies
- **zizmor**
  - GitHub Actions security linting
- **SHA-pinned Actions**
  - no tag-based references
- **uv.lock + hashes**
  - deterministic, tamper-evident installs
- **7-day age gate**
  - never install packages < 7 days old
- **SSH signature verification**
  - signed commits and model downloads

# Agentic Security

Supply chain security is not so different:

- Agents run `npx`, `pip install`, `npm install` - all bypass your lockfile
- Agents add dependencies without checking age, provenance, or equivalents
- Unsupervised/automated CI pulling backdoored packages is worse with agents

**However**, LLMs introduce a whole new class of attacks — the **Lethal Trifecta**:

- **Access to private data** — email, files, credentials, calendar, contacts
- **Exposure to untrusted content** — web pages, incoming email, documents, API responses
- **Ability to take consequential actions** — send messages, write files, execute commands, make purchases

Further reading:

- **Agentic Security Research** (https://github.com/lhl/agentic-security)
- **OpenClaw is Amazing, but... Don't Install It** (https://github.com/shisa-ai/shisad/blob/main/docs/analysis/openclaw-awesome-but-dontinstall.md)

# Resources

**Harness engineering & dark factory:**

- Lopopolo: https://openai.com/index/harness-engineering/
  - Latent Space: https://latent.space/p/harness-eng
  - OpenAI Symphony: https://github.com/openai/symphony
- StrongDM Software Factories: https://factory.strongdm.ai/

**My projects & workflow:**

- shisad: https://github.com/shisa-ai/shisad
  - agentic-security: https://github.com/lhl/agentic-security
  - agentic-memory: https://github.com/lhl/agentic-memory
- Workflow writeup: https://agenticcodingweekly.com (ACW #14)

**Supply chain security:**

  - https://gist.github.com/lhl/